



Name: Gourav Gupta

B.Tech : Computer Science & Engineering (C.S.E.)

Roll number : 23CS8041

Gender : Male

Date of Birth : 09-10-2004

Category : OBC

E-mail : gouravgupta76058@gmail.com

Contact : 7605899433

Address : 69, Picnic Garden Road, Kolkata-700039



EDUCATIONAL QUALIFICATION:

• Graduation

SEM	1	2	3	4	5	6	7	8
SGPA	8.57	8.60	8.54	8.18	8.70	9.04		
CGPA	8.57	8.58	8.57	8.47	8.52	8.61		

• Higher Secondary & Secondary

Year	Degree/Examination	Institution/Board	Percentage
2022	Class XII	Central Board of Secondary Education	88.6%
2020	Class X	Central Board of Secondary Education	92.8%

ACHIEVEMENTS:

• Competitive Programming & Coding Proficiency

- **Codeforces Expert** : Attained an **Expert** rank (Max Rating: **1626**). Profile: [gourav_27](#).
- **CodeChef 4-Star** : Reached **4-Star** status (Max Rating: **1880**). Profile: [gourav_2027](#).
- **LeetCode Knight** : (**Top 4%** globally) with a peak rating of **1889**. Profile: [Gourav gupta 2004](#).
- **Problem Solving**: Successfully cleared **1700+** algorithmic challenges across platforms including AtCoder, CSES, and TakeUForward.

• Notable Contest Rankings

- Ranked **197th** globally in CodeChef Starters 173 among 47,800+ competitors.
- Placed **771st** and **837th** globally in Codeforces Rounds 1072 and 1071 respectively, each with 40,000+ participants.

• Amazon HackOn 6.0 Finalist: Ranked among the **Top 29 teams out of 70000+ nationwide candidates**.

• Academic Achievements:

- **WBJEE 2023**: Secured a state rank of **1212** out of **100,000** candidates.
- **JEE Mains 2023**: Achieved an All India Rank of **17795** among **1.17 million** aspirants.

TECHNICAL SKILLS:

- **Technical Skills** : Data Structures and Algorithms(DSA), Object-Oriented Programming(OOPS), Operating Systems(OS), Database Management Systems(DBMS), Computer Networks(CN), Machine Learning(ML), Supervised & Unsupervised Learning, Neural Networks(ANN,CNN,RNN),LSTM, Transformers, Generative AI (Gen AI) .
- **Non-Technical Skills** : Problem Solving, Team Leadership, Project Management, Communication, Decision Making, Analytical Thinking, Creative Thinking, Time Management.
- **Languages** : C, C++, Java, Python, SQL, HTML, CSS, Javascript.
- **Technologies/Frameworks** : Redis, React.js, Node.js, Express.js, Next.js , MongoDB, MySQL, NumPy, Pandas, LangChain.
- **Development Tools** : Git, Github, Visual Studio Code, Sublime text, Google Colab, Jupyter Notebook.

INTERESTS & TECHNICAL PROFICIENCIES:

- ❖ **LLD:** Proficient in design patterns – **Creational** (Singleton, Factory Method, Abstract Factory, Builder), **Structural** (Adapter, Bridge, Composite, Decorator, Facade), **Behavioral** (Chain of Responsibility, Command, Iterator, Mediator, Memento, Observer, Strategy, Template Method, Visitor).
- ❖ **HLD/System Design:** Load balancing, database sharding & replication, caching, CDNs, message queues, event-driven & microservices architecture, consistent hashing, rate limiting, CAP theorem.

PROJECTS:

RAG PDF ChatBot : [Repo link](#)

- ❖ Built a **full-stack RAG**-based PDF chatbot using **Next.js** and **Node.js**, enabling context-aware chat over uploaded documents via natural language queries.
- ❖ Implemented semantic search using **vector embeddings** with **LangChain**, **Gemini API**, and **Qdrant** to retrieve relevant document chunks and improve answer accuracy.
- ❖ Designed **asynchronous** job processing using **BullMQ** and **Valkey** for scalable, non-blocking PDF extraction and embedding generation.
- ❖ **Containerized services** with **Docker** to orchestrate **Qdrant** and **Valkey**, enabling one-command setup across environments.
- ❖ Tech Stack: **Next.js**, **React**, **Shadcn UI**, **Node.js**, **LangChain**, **Gemini API**, **BullMQ**, **Qdrant**, **Valkey**, **Docker**, **Clerk**.

Echo : [Repo link](#)

- ❖ Built a **real-time Full-stack** chat application enabling users to send and receive instant messages. With online/offline status updates for enhanced interactivity.
- ❖ Designed a scalable MongoDB schema to store users, chat rooms, and messages with efficient retrieval and relationship handling.
- ❖ Leveraged **Socket.io** for **real-time, Full-Duplex communication** between clients and server. Implemented user **authentication**, and secure communication using **JWT and RESTful APIs**.
- ❖ Future enhancements include media sharing (images, videos), message reactions, and read receipts for a more engaging user experience.
- ❖ Tech Stack: **React.js**, **Node.js**, **Express.js**, **MongoDB**, **Socket.io**, **JWT**.

PubSub LLD : [Repo link](#)

- ❖ Designed a low-level Pub-Sub messaging system in Java using **Observer**, **Repository**, and **layered architecture** patterns to enable topic-based message delivery.
- ❖ Implemented **dual-channel notification system** with Email and Realtime observers, dynamically adjusting delivery based on subscriber online/offline status.
- ❖ Built **thread-safe, non-blocking message fan-out** using **CompletableFuture** and **CopyOnWriteArrayList** for **asynchronous, concurrent subscriber notification**.
- ❖ Developed **repository layer** using **ConcurrentHashMap**-backed interfaces, following **Dependency Inversion Principle** for clean separation of business logic and data access.
- ❖ Tech Stack: **Java**, **Observer Pattern**, **Repository Pattern**, **Layered Architecture**, **CompletableFuture**, **ConcurrentHashMap**.

EXPERIENCE:

REcursion, NIT Durgapur | Senior Coordinator

Jan 2025 – Present

- ❖ **Instructed** Competitive Programming fundamentals to **Sophomore & Fresher** students via **bi-weekly sessions**, directly **mentoring 8 students** to strengthen algorithmic logic and coding proficiency.
- ❖ Managed quarterly coding contests (**200+** avg. participants) and facilitated **post-contest analyses** to resolve technical queries and improve participant performance.
- ❖ Authored multiple algorithmic problems as **Problem Setter** for club-level competitions, including including [RECode 31C](#) and [RECode 32E](#).